z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

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Question 1

Let X have a normal distribution with mean 10 and variance 36. Find

1. $P(X > 5)$;
2. $P(4 < X < 16)$;
3. $P(X < 8)$;
4. $P(X < 20)$;
5. $P(X > 16)$.

$\mu = 10 \qquad \sigma^2 = 36$

Z-table

$$\frac{X - \mu}{\sigma}$$

$P(X \geq a)$

$= 1 - P(X \leq a)$

$= 1 - P\left(\frac{X-\mu}{\sigma} \leq \frac{a-\mu}{\sigma}\right)$

$a < X < b$

$\frac{a-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}$

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Solution

- We are given $X \sim N(\mu, \sigma^2)$ with $\mu = 10$ and $\sigma^2 = 36$.

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Solution

- We are given $X \sim N(\mu, \sigma^2)$ with $\mu = 10$ and $\sigma^2 = 36$.
- Since standard deviation is the positive square root of variance,

$$\sigma = \sqrt{36} = 6.$$

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Solution

- We are given $X \sim N(\mu, \sigma^2)$ with $\mu = 10$ and $\sigma^2 = 36$.
- Since standard deviation is the positive square root of variance,

$$\sigma = \sqrt{36} = 6.$$

- Standardize any threshold a using

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 10}{6} \sim N(0, 1), \qquad P(X < a) = \Phi\left(\frac{a - 10}{6}\right).$$

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Solution

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- Standardize any threshold a using

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 10}{6} \sim N(0, 1), \qquad P(X < a) = \Phi\left(\frac{a - 10}{6}\right).$$

- Here $\Phi(z) = P(Z \leq z)$, and for a continuous variable $P(Z = z) = 0$.

$P(Z > z) = 1 - P(Z \leq z)$

$= 1 - \Phi(z)$

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- (a) For $X > 5$,

$$P(X > 5) = P\left(Z > \frac{5-10}{6}\right) = P(Z > -5/6).$$

$$= 1 - \Phi(-0.8333) = \Phi(0.8333) \approx \boxed{0.7977}.$$

$\Phi(-z) = 1 - \Phi(z)$

$\Phi(z) = P(Z \le z)$

$P(Z > -5/6) = 1 - P(Z \le -5/6)$

$= 1 - \Phi(-0.833)$

$= \Phi(0.833)$

$\Phi(-z) = 1 - \Phi(z)$

$\Phi(z) = 1 - \Phi(-z)$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

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- (a) For $X > 5$,

$$P(X > 5) = P\left(Z > \frac{5-10}{6}\right) = P(Z > -5/6).$$

$$= 1 - \Phi(-0.8333) = \Phi(0.8333) \approx \boxed{0.7977}.$$

- (b) For $4 < X < 16$, compute both standardized endpoints:

$$z_L = \frac{4-10}{6} = -1, \qquad z_U = \frac{16-10}{6} = 1.$$

$$P(4 < X < 16) = \Phi(1) - \Phi(-1)$$

$$= 0.8413 - 0.1587 \approx \boxed{0.6827}.$$

$P(-1 < Z < 1)$

$\Phi(1) - \Phi(-1)$

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- (a) For $X > 5$,

$$P(X > 5) = P\left(Z > \frac{5-10}{6}\right) = P(Z > -5/6).$$

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- (b) For $4 < X < 16$, compute both standardized endpoints:

$$z_L = \frac{4-10}{6} = -1, \qquad z_U = \frac{16-10}{6} = 1.$$

$$P(4 < X < 16) = \Phi(1) - \Phi(-1)$$

$$= 0.8413 - 0.1587 \approx \boxed{0.6827}.$$

- (c) Standardize 8:

$$P(X < 8) = \Phi\left(\frac{8-10}{6}\right) = \Phi(-1/3) \approx \boxed{0.3694}.$$

= 1 - Φ(1/3) =

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- (d) Standardize 20:

$$P(X < 20) = \Phi\left(\frac{20 - 10}{6}\right) = \Phi(5/3) \approx \boxed{0.9522}.$$

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- (d) Standardize 20:

$$P(X < 20) = \Phi\left(\frac{20-10}{6}\right) = \Phi(5/3) \approx \boxed{0.9522}.$$

- (e) For the upper tail, use the complement:

$$P(X > 16) = 1 - \Phi\left(\frac{16-10}{6}\right) = 1 - \Phi(1) = 1 - 0.8413 \approx \boxed{0.1587}.$$

$= 1 - P(X < 16)$

$= 1 - P\left(Z < \frac{16-10}{6}\right)$

$= 1 - \Phi(1)$

$= 1 - 0.8413$

≈ 0.1587

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Question 2

A bolt manufacturer specifies acceptable diameters from 1.19 to 1.21 inches. Produced bolt diameters are normally distributed with mean 1.20 inches and standard deviation 0.005 inches. What percentage of produced bolts falls outside specifications?

$N(\mu, \sigma^2)$ $\quad \mu = 1.20$

$\sigma = 0.005$

Let diam be x

Acceptable if $\quad 1.19 \leq x \leq 1.21$

What $P(\text{Accept})$ $\quad P(1.19 \leq x \leq 1.21)$

$1 - P(1.19 \leq x \leq 1.21)$

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Solution

- Define bolt diameter $X \sim N(1.20, 0.005^2)$.

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Solution

- Define bolt diameter $X \sim N(1.20, 0.005^2)$.
- A bolt is unacceptable if $X < 1.19$ or $X > 1.21$.

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Solution

- Define bolt diameter $X \sim N(1.20, 0.005^2)$.
- A bolt is unacceptable if $X < 1.19$ or $X > 1.21$.
- The standardized lower specification is

$$z_L = \frac{1.19 - 1.20}{0.005} = \frac{-0.01}{0.005} = -2.$$

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Solution

- Define bolt diameter $X \sim N(1.20, 0.005^2)$.
- A bolt is unacceptable if $X < 1.19$ or $X > 1.21$.
- The standardized lower specification is

$$z_L = \frac{1.19 - 1.20}{0.005} = \frac{-0.01}{0.005} = -2.$$

- The standardized upper specification is

$$z_U = \frac{1.21 - 1.20}{0.005} = \frac{0.01}{0.005} = 2.$$

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Solution

- Define bolt diameter $X \sim N(1.20, 0.005^2)$.
- A bolt is unacceptable if $X < 1.19$ or $X > 1.21$.
- The standardized lower specification is

$$z_L = \frac{1.19 - 1.20}{0.005} = \frac{-0.01}{0.005} = -2.$$

- The standardized upper specification is

$$z_U = \frac{1.21 - 1.20}{0.005} = \frac{0.01}{0.005} = 2.$$

- The two tails are disjoint, so add their probabilities:

$$P(\text{outside}) = P(X < 1.19) + P(X > 1.21).$$

$$= \Phi(-2) + [1 - \Phi(2)].$$

$1 - P(1.19 \le X \le 1.21)$

$1 - P(-2 \le Z \le 2)$

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- By symmetry $1 - \Phi(2) = \Phi(-2)$:

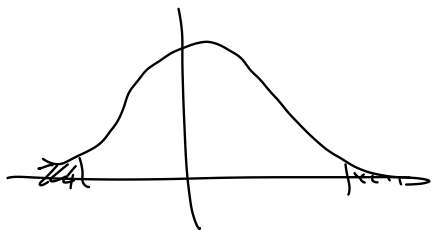
$1 - \Phi(z) = \Phi(-z)$

$$P(\text{outside}) = 2\Phi(-2) = 2(0.0227501)$$

$$\approx \boxed{0.04550}.$$

$= 2\left[1 - \Phi(2)\right]$

$= 0.04550$



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- By symmetry $1 - \Phi(2) = \Phi(-2)$:

$$P(\text{outside}) = 2\Phi(-2) = 2(0.0227501)$$

$$\approx \boxed{0.04550}.$$

- Converting a fraction to a percentage:

$$0.04550 \times 100\% \approx \boxed{4.55\%}.$$

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Question 3

Weekly demand for a product is normal with mean 1000 and standard deviation 200. There are 2200 units on hand, and no replenishment will arrive during the next two weeks. Assume demands in different weeks are independent.

1. Find the probability that demand is less than 1100 in *each* of the next two weeks.
2. Find the probability that *total* two-week demand exceeds 2200.

$X \rightarrow$ Demand

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Solution

- Let D_1, D_2 denote demand in weeks 1 and 2. The model is

$$D_i \sim N(1000, 200^2), \qquad D_1, D_2. \text{ are independent}$$

$\mu = 1000$ $\qquad$ $\sigma = 200$

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Solution

- Let D_1, D_2 denote demand in weeks 1 and 2. The model is

$$D_i \sim N(1000, 200^2), \qquad D_1, D_2. \text{ are independent}$$

- (a) First compute the probability for one week:

$$P(D_i < 1100) = \Phi\left(\frac{1100 - 1000}{200}\right) = \Phi(0.5).$$

$$\Phi(0.5) \approx 0.691462.$$

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Solution

- Let D_1, D_2 denote demand in weeks 1 and 2. The model is

$$D_i \sim N(1000, 200^2), \qquad D_1, D_2. \text{ are independent}$$

- (a) First compute the probability for one week:

$$P(D_i < 1100) = \Phi\Big(\frac{1100 - 1000}{200}\Big) = \Phi(0.5).$$

$$\Phi(0.5) \approx 0.691462.$$

- By independence, multiply the two probabilities:

$$P(D_1 < 1100, D_2 < 1100) = (0.691462)^2 \approx \boxed{0.4781}.$$

$P(D_1 < 1100) \times P(D_2 < 1100) = \Phi(0.5) \times \Phi(0.5)$

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Solution

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- (b) Form the two-week total $T = D_1 + D_2$.

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- (b) Form the two-week total $T = D_1 + D_2$. → Total demand → $T = D_1 + D_2$

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- Add means, and (because of independence) add variances:

$$E[T] = 1000 + 1000 = 2000,$$

$$\mathrm{Var}(T) = 200^2 + 200^2 = 80000.$$

$\mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2]$

$T \sim N(\mu_T, \sigma_T^2)$

μ

$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2)$

$\sigma^2 = \sigma_1^2 + \sigma_2^2$

$= 200^2 + 200^2$

$= 80000$

$\sigma = \mathrm{sqrt}(80000)$

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- Add means, and (because of independence) add variances:

$$E[T] = 1000 + 1000 = 2000,$$

$$\text{Var}(T) = 200^2 + 200^2 = 80000.$$

- Thus $T \sim N(2000, 80000)$ and $\sigma_T = \sqrt{80000} = 200\sqrt{2}$.

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- Add means, and (because of independence) add variances:

$$E[T] = 1000 + 1000 = 2000,$$

$$\mathrm{Var}(T) = 200^2 + 200^2 = 80000.$$

- Thus $T \sim N(2000, 80000)$ and $\sigma_T = \sqrt{80000} = 200\sqrt{2}$.
- Now standardize the inventory threshold:

$$P(T > 2200) = 1 - \Phi\left(\frac{2200 - 2000}{200\sqrt{2}}\right)$$

$$= 1 - \Phi(0.7071) \approx 1 - 0.7602 = \boxed{0.2398}.$$

$D_1 + D_2 > 2200$

$1 - P(T < 2200)$

$1 - \Phi(\frac{1}{\sqrt{2}})$

$\frac{T - \mu_T}{\sigma_T} = \frac{T - 2000}{200\sqrt{2}}$

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Question 4

Let $U_1, \ldots, U_{16}$ be independent random variables, each uniform on $(0, 1)$. Approximate the probability that $U_1 + \cdots + U_{16}$ exceeds 10.

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Solution

- For each $U_i \sim \text{Uniform}(0,1)$, the density is $f(u) = 1$ for $0 < u < 1$.

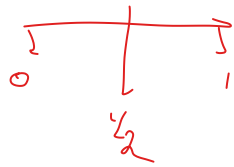
$f(u) = 1$

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Solution

- For each $U_i \sim \text{Uniform}(0,1)$, the density is $f(u) = 1$ for $0 < u < 1$.
- Compute its mean using integration:

$$E[U_i] = \int_0^1 u\, du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}.$$



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Solution

- For each $U_i \sim \text{Uniform}(0, 1)$, the density is $f(u) = 1$ for $0 < u < 1$.
- Compute its mean using integration:

$$E[U_i] = \int_0^1 u \, du = \left[\frac{u^2}{2} \right]_0^1 = \frac{1}{2}.$$

- Compute its second moment:

$$E[U_i^2] = \int_0^1 u^2 \, du = \left[\frac{u^3}{3} \right]_0^1 = \frac{1}{3}.$$

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Solution

- For each $U_i \sim \text{Uniform}(0,1)$, the density is $f(u) = 1$ for $0 < u < 1$.
- Compute its mean using integration:

$$E[U_i] = \int_0^1 u\, du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}.$$

- Compute its second moment:

$$E[U_i^2] = \int_0^1 u^2\, du = \left[\frac{u^3}{3}\right]_0^1 = \frac{1}{3}.$$

- Therefore $\text{Var}(U_i) = 1/3 - (1/2)^2 = 1/12$.
- Define $S_{16} = \sum_{i=1}^{16} U_i$. By linearity,

$$E[S_{16}] = 16\left(\frac{1}{2}\right) = 8.$$

$U_1 + U_2 + \cdots U_{16} < 10$

S_{16}

$E[U_1 + \cdots U_{16}] = E[U_1] + \cdots E[U_{16}] = 16 \times \frac{1}{2} = 8$

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Solution

- For each $U_i \sim \text{Uniform}(0,1)$, the density is $f(u) = 1$ for $0 < u < 1$.
- Compute its mean using integration:

$$E[U_i] = \int_0^1 u \, du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}.$$

- Compute its second moment:

$$E[U_i^2] = \int_0^1 u^2 \, du = \left[\frac{u^3}{3}\right]_0^1 = \frac{1}{3}.$$

- Therefore $\text{Var}(U_i) = 1/3 - (1/2)^2 = 1/12$.
- Define $S_{16} = \sum_{i=1}^{16} U_i$. By linearity,

$$E[S_{16}] = 16\left(\frac{1}{2}\right) = 8.$$

- Independence lets us sum the variances:

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$$\left(\frac{1}{12}\right) \quad \frac{4}{}$$

- Apply the Central Limit Theorem (approximately):

$$\frac{S_{16} - 8}{\sqrt{4/3}} \approx N(0, 1).$$

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- Apply the Central Limit Theorem (approximately):

$$\frac{S_{16} - 8}{\sqrt{4/3}} \approx N(0,1).$$

$$\frac{S_{16} - 8}{\sqrt{4/3}} \sim N(0,1)$$

- Standardize and evaluate:

$$P(S_{16} > 10) \approx 1 - \Phi\left(\frac{10 - 8}{\sqrt{4/3}}\right)$$

$$= 1 - \Phi(\sqrt{3}) \approx \boxed{0.0416}.$$

$$P(S_{16} > 10)$$

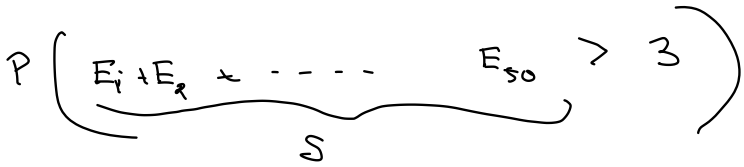
$$1 - P(S_{16} \leq 10)$$

$$1 - P\left(\frac{S_{16} - \mu}{\sigma} \leq \frac{10 - \mu}{\sigma}\right)$$

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Question 5

Fifty real numbers are independently rounded to the nearest integer and then added. Treat the individual rounding errors as independent uniforms on $(-0.5, 0.5)$. Approximate the probability that the rounded sum differs from the true sum by more than 3.



$E_1 + E_2 + \cdots$

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Solution

- Let E_i be one rounding error, with $E_i \sim \text{Uniform}(-1/2, 1/2)$.

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Solution

- Let E_i be one rounding error, with $E_i \sim \text{Uniform}(-1/2, 1/2)$.
- By symmetry, $E[E_i] = 0$. By the uniform variance formula,

$$\text{Var}(E_i) = \frac{(b-a)^2}{12} = \frac{(1)^2}{12} = \frac{1}{12}.$$

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Solution

- Let E_i be one rounding error, with $E_i \sim \text{Uniform}(-1/2, 1/2)$.
- By symmetry, $E[E_i] = 0$. By the uniform variance formula,

$$\text{Var}(E_i) = \frac{(b-a)^2}{12} = \frac{(1)^2}{12} = \frac{1}{12}.$$

- The total rounding error is $S = \sum_{i=1}^{50} E_i$.

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Solution

- Let E_i be one rounding error, with $E_i \sim \text{Uniform}(-1/2, 1/2)$.
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- The total rounding error is $S = \sum_{i=1}^{50} E_i$.
- Because the errors are independent,

$$E[S] = 50(0) = 0, \quad \text{Var}(S) = 50/12, \quad \sigma_S = \sqrt{50/12}.$$

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Solution

- Let E_i be one rounding error, with $E_i \sim \text{Uniform}(-1/2, 1/2)$.
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$$E[S] = 50(0) = 0, \quad \text{Var}(S) = 50/12, \quad \sigma_S = \sqrt{50/12}.$$

- The error exceeds 3 in magnitude precisely when $S < -3$ or $S > 3$:

$$P(|S| > 3) = P(S < -3) + P(S > 3).$$

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Solution

- Let E_i be one rounding error, with $E_i \sim \text{Uniform}(-1/2, 1/2)$.
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- The error exceeds 3 in magnitude precisely when $S < -3$ or $S > 3$:

$$P(|S| > 3) = P(S < -3) + P(S > 3).$$

- Apply the CLT to the sum:

$$Z = \frac{S - 0}{\sqrt{50/12}} \approx N(0, 1).$$

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- The standardized threshold is

$$z_0 = \frac{3}{\sqrt{50/12}} \approx 1.46969.$$

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- The standardized threshold is

$$z_0 = \frac{3}{\sqrt{50/12}} \approx 1.46969.$$

- Symmetry gives

$$P(|S| > 3) \approx 2[1 - \Phi(1.46969)]$$
$$\approx 2(0.07082) = \boxed{0.1416}.$$

$\phi(z) = 1 - \phi(-z)$

$P(Z \geq z) = 1 - P(Z \leq z) = 1 - \phi(z)$

$P(|S| > 3) = P(S < -3) + P(S > 3)$

$P\left(S < \frac{-3}{\sqrt{50/12}}\right) + P\left(S > \frac{3}{\sqrt{50/12}}\right)$

$2[1 - \phi(1.46969)]$

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Question 6:

Let $X_1, \ldots, X_n$ be a random sample from a normal population with mean μ and variance σ^2.

Assume that μ is known.

1. Determine the maximum likelihood estimator of σ^2.
2. Find the expected value of this estimator.

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Solution

- The density of each observation is

$$f(x_i|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right).$$

Given a density fn

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Solution

- The density of each observation is

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

- By independence, the likelihood function is

$$L(\sigma^2) = \prod_{i=1}^{n} f(x_i; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right).$$

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Solution

- The density of each observation is

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- By independence, the likelihood function is

$$L(\sigma^2) = \prod_{i=1}^{n} f(x_i; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right).$$

- Taking the natural logarithm,

$$\ell(\sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$



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Solution

- The density of each observation is

$$f(x_i;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right).$$

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- Taking the natural logarithm,

$$\ell(\sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

- Let $v = \sigma^2 > 0$. In terms of v, the log-likelihood is

$$\ell(v) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log v - \frac{1}{2v}\sum_{i=1}^{n}(x_i-\mu)^2.$$

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- Differentiating with respect to v,

$$\frac{d\ell}{dv} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

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- Differentiating with respect to v,

$$\frac{d\ell}{dv} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

- Setting $d\ell/dv = 0$ and multiplying by $2v^2$:

$$-nv + \sum_{i=1}^{n}(x_i - \mu)^2 = 0 \quad \Longrightarrow \quad v = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2.$$

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$$-nv + \sum_{i=1}^{n}(x_i - \mu)^2 = 0 \quad \Longrightarrow \quad v = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2.$$

- Hence, the maximum likelihood estimator is

$$\boxed{\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2.}$$

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- To verify that the stationary point is a maximum, calculate the second derivative:

$$\frac{d^2\ell}{dv^2} = \frac{n}{2v^2} - \frac{1}{v^3}\sum_{i=1}^{n}(x_i - \mu)^2.$$

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$$\frac{d^2\ell}{dv^2} = \frac{n}{2v^2} - \frac{1}{v^3}\sum_{i=1}^{n}(x_i - \mu)^2.$$

- At the stationary point, $\sum_{i=1}^{n}(x_i - \mu)^2 = nv$, which gives

$$\frac{d^2\ell}{dv^2} = \frac{n}{2v^2} - \frac{nv}{v^3} = -\frac{n}{2v^2} < 0.$$

$v = \frac{1}{n}\sum (x_i - \mu)^2$

$\sum (x_i - \mu)^2 = nv$

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- Since the second derivative is strictly negative, the stationary point is a strict local maximum.

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$$\frac{d^2\ell}{dv^2} = \frac{n}{2v^2} - \frac{nv}{v^3} = -\frac{n}{2v^2} < 0.$$

- Since the second derivative is strictly negative, the stationary point is a strict local maximum.
- Hence this stationary point is the unique global maximum.

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- We now evaluate the expectation of $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$.

$$\mathbb{E}\left[\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i}^{n}(x_i - \mu)^2\right]$$

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- We now evaluate the expectation of $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$.
- By linearity of expectation,

$$\mathbb{E}[\widehat{\sigma}^2] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[(X_i - \mu)^2].$$

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2\right] = \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(X_i-\mu)^2\right] \quad \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i-\mu]^2$$

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- We now evaluate the expectation of $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$.
- By linearity of expectation,

$$\mathbb{E}[\widehat{\sigma}^2] = \frac{1}{n}\sum_{i=1}^{n}\underbrace{\mathbb{E}[(X_i - \mu)^2]}_{\mathrm{Var}(X_i)}.$$

$X_i \sim N(\mu, \sigma^2)$

- Since $\mu = \mathbb{E}[X_i]$ is the true known mean,

$$\mathbb{E}[(X_i - \mu)^2] = \mathrm{Var}(X_i) = \sigma^2.$$

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$$\mathbb{E}[(X_i - \mu)^2] = \mathsf{Var}(X_i) = \sigma^2.$$

- Substituting this back,

$$\mathbb{E}[\widehat{\sigma}^2] = \frac{1}{n}\sum_{i=1}^{n}\sigma^2 = \frac{n\sigma^2}{n} = \boxed{\sigma^2.}$$

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- We now evaluate the expectation of $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$.
- By linearity of expectation,

$$\mathbb{E}[\widehat{\sigma}^2] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[(X_i - \mu)^2].$$

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Question 7:

Let $X_1, X_2, \ldots, X_n$ be an independent and identically distributed (i.i.d.) random sample from an exponential distribution with unknown rate parameter $\lambda > 0$, whose probability density function is given by:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Determine the maximum likelihood estimator (MLE) of λ.

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Solution

- The likelihood function $L(\lambda)$ for the sample $X_1, \ldots, X_n$ is:

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda X_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} X_i\right)$$

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Solution

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Step 1

Write the likelihood

- Taking the natural logarithm yields the log-likelihood:

$$\ell(\lambda) = \ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} X_i$$

Step 2

Derivative = 0

$$\frac{d\ell}{d\lambda} = 0$$

Solve for the parameter.

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Solution

- The likelihood function $L(\lambda)$ for the sample $X_1, \ldots, X_n$ is:

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda X_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} X_i\right)$$

- Taking the natural logarithm yields the log-likelihood:

$$\ell(\lambda) = \ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} X_i$$

- Differentiating with respect to λ and setting the derivative to zero:

$$\frac{d}{d\lambda}\ell(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} X_i = 0$$

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Solution

- The likelihood function $L(\lambda)$ for the sample $X_1, \ldots, X_n$ is:

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda X_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} X_i\right)$$

- Taking the natural logarithm yields the log-likelihood:

$$\ell(\lambda) = \ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} X_i$$

- Differentiating with respect to λ and setting the derivative to zero:

$$\frac{d}{d\lambda}\ell(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} X_i = 0$$

- $$\implies \boxed{\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} X_i} = \frac{1}{\overline{X}}}$$

$\frac{\sum X_i}{n} = \overline{X}$

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- To verify that this point is a maximum, check the second derivative:

$$\frac{d^2}{d\lambda^2}\ell(\lambda) = -\frac{n}{\lambda^2} < 0 \qquad \text{for all } \lambda > 0$$

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- To verify that this point is a maximum, check the second derivative:

$$\frac{d^2}{d\lambda^2}\ell(\lambda) = -\frac{n}{\lambda^2} < 0 \qquad \text{for all } \lambda > 0$$

- Thus, $\hat{\lambda}_{\mathrm{MLE}} = \frac{1}{\bar{X}}$ uniquely maximizes the likelihood.

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Question 8

A six-sided die, in which each side is equally likely to appear, is repeatedly rolled until the total of all rolls exceeds 400.

Approximate the probability that this will require more than 140 rolls.

$X_1 \in \{1 \dots 6\}$

$X_2 \in \{1 \dots 6\}$

$X_1 + X_2 + \dots + X_n > 400$

$X_1 + \dots + X_{140} \leq 400$

Then you need more than 140 rolls.

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Solution

- Let X_i denote the outcome of the i-th roll. Then $X_1, \ldots, X_{140}$ are i.i.d. random variables with

$$P(X_i = k) = \frac{1}{6}, \qquad k = 1, \ldots, 6.$$

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Solution

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$$P(X_i = k) = \frac{1}{6}, \qquad k = 1, \ldots, 6.$$

- Compute the mean of each roll:

$$E[X_i] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2} = 3.5.$$

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Solution

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- Compute the mean of each roll:

$$E[X_i] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2} = 3.5.$$

- Compute its second moment:

$$E[X_i^2] = \frac{1^2 + 2^2 + \cdots + 6^2}{6} = \frac{91}{6}.$$

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Solution

- Let X_i denote the outcome of the i-th roll. Then $X_1, \ldots, X_{140}$ are i.i.d. random variables with

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- Compute its second moment:

$$E[X_i^2] = \frac{1^2 + 2^2 + \cdots + 6^2}{6} = \frac{91}{6}.$$

- Therefore,

$$\mathrm{Var}(X_i) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}.$$

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- Define $S_{140} = \sum_{i=1}^{140} X_i$. By linearity,

$$E[S_{140}] = 140(3.5) = 490.$$

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- Define $S_{140} = \sum_{i=1}^{140} X_i$. By linearity,

$$E[S_{140}] = 140(3.5) = 490.$$

$E[X_i] = 3.5$

- Independence lets us sum the variances:

$$\mathrm{Var}(S_{140}) = 140\left(\frac{35}{12}\right) = \frac{1225}{3}.$$

$$\mathrm{SD}(S_{140}) = \sqrt{1225/3}.$$

$\sigma^2 = \frac{35}{12}$

$\sigma = \sqrt{\frac{1225}{3}}$

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- More than 140 rolls are required precisely when $S_{140} \leq 400$.

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- More than 140 rolls are required precisely when $S_{140} \leq 400$.
- Apply the Central Limit Theorem (approximately):

$$\frac{S_{140} - 490}{\sqrt{1225/3}} \approx N(0,1).$$

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- More than 140 rolls are required precisely when $S_{140} \leq 400$.
- Apply the Central Limit Theorem (approximately):

$$\frac{S_{140} - 490}{\sqrt{1225/3}} \approx N(0,1).$$

- Since S_{140} is integer-valued, use the continuity correction and standardize:

$$P(S_{140} \leq 400) \approx \Phi\left(\frac{400.5 - 490}{\sqrt{1225/3}}\right).$$

$$P\left(\frac{S_{140} - 490}{\sqrt{1225/3}} < \frac{400 - 490}{\sqrt{1225/3}}\right)$$

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- Define $S_{140} = \sum_{i=1}^{140} X_i$. By linearity,

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$1 - \Phi(-\ldots)$ → Refer to z table.

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Question 9

Let $X_1, \ldots, X_n$ be a random sample from the distribution whose density function is

$$f(x;\theta) = \begin{cases} e^{-(x-\theta)}, & x \geq \theta, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the maximum likelihood estimator of θ.

$f(x_1 \cdots x_n) = f(x_1;\theta) f(x_2;\theta) \cdots$

If $\theta >$ some x_i

$f(x_1 \cdots) \cdots \times 0$

$= 0$

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- Since $X_1, \ldots, X_n$ are independent, the joint density is the product of the individual densities:

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta).$$

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- Since $X_1, \ldots, X_n$ are independent, the joint density is the product of the individual densities:

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta).$$

- Substituting the given density with indicator functions,

$$L(\theta) = \prod_{i=1}^{n} e^{-(x_i - \theta)} \mathbf{1}_{\{x_i \geq \theta\}}.$$

$$f = \begin{cases} e^{-(x_i - \theta)} & \text{if } x_i \geq \theta \\ 0 & \text{elsewhere.} \end{cases}$$

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- Substituting the given density with indicator functions,

$$L(\theta) = \prod_{i=1}^{n} e^{-(x_i - \theta)} \mathbf{1}_{\{x_i \geq \theta\}}.$$

- The product of exponential terms becomes

$$\prod_{i=1}^{n} e^{-(x_i - \theta)} = \exp\left(- \sum_{i=1}^{n} x_i + n\theta \right).$$

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$$L(\theta) = \prod_{i=1}^{n} e^{-(x_i - \theta)} \mathbf{1}_{\{x_i \geq \theta\}}.$$

- The product of exponential terms becomes

$$\prod_{i=1}^{n} e^{-(x_i - \theta)} = \exp\left(-\sum_{i=1}^{n} x_i + n\theta \right).$$

- All observations must satisfy $x_i \geq \theta$ simultaneously:

$$x_i \geq \theta \quad \forall i \quad \Longleftrightarrow \quad \theta \leq \min_{1 \leq i \leq n} x_i = x_{(1)}.$$

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- Therefore, the likelihood function is

$$L(\theta) = \begin{cases} \exp\left(-\sum_{i=1}^{n} x_i + n\theta\right), & \theta \leq x_{(1)}, \\ 0, & \theta > x_{(1)}, \end{cases}$$

where $x_{(1)} = \min\{x_1, \ldots, x_n\}$.

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- Therefore, the likelihood function is

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  where $x_{(1)} = \min\{x_1, \ldots, x_n\}$.

- In the feasible region $\theta \leq x_{(1)}$, the log-likelihood is

$$\ell(\theta) = -\sum_{i=1}^{n} x_i + n\theta.$$

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- In the feasible region $\theta \leq x_{(1)}$, the log-likelihood is

$$\ell(\theta) = -\sum_{i=1}^{n} x_i + n\theta.$$

- Differentiating with respect to θ,

$$\frac{d\ell}{d\theta} = n > 0.$$

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- Therefore, the likelihood function is

$$L(\theta) = \begin{cases} \exp\left(-\sum_{i=1}^{n} x_i + n\theta\right), & \theta \leq x_{(1)}, \\ 0, & \theta > x_{(1)}, \end{cases}$$

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- In the feasible region $\theta \leq x_{(1)}$, the log-likelihood is

$$\ell(\theta) = -\sum_{i=1}^{n} x_i + n\theta.$$

- Differentiating with respect to θ,

$$\frac{d\ell}{d\theta} = n > 0.$$

- Thus $\ell(\theta)$ is strictly increasing in θ on $(-\infty, x_{(1)}]$.

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- Therefore, the likelihood function is

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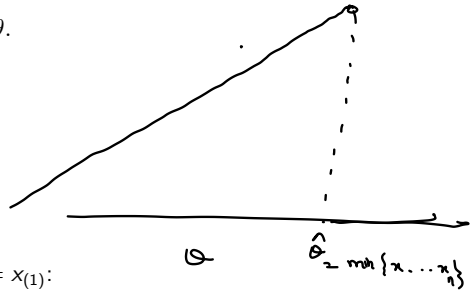
where $x_{(1)} = \min\{x_1, \dots, x_n\}$.

- In the feasible region $\theta \le x_{(1)}$, the log-likelihood is

$$\ell(\theta) = -\sum_{i=1}^{n} x_i + n\theta.$$

- Differentiating with respect to θ,

$$\frac{d\ell}{d\theta} = n > 0.$$



- Thus $\ell(\theta)$ is strictly increasing in θ on $(-\infty, x_{(1)}]$.
- The maximum is attained at the largest feasible value, $\theta = x_{(1)}$:

$$\widehat{\theta}_{\mathrm{MLE}} = X_{(1)} = \min\{X_1, \dots, X_n\}.$$

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Question 10

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. random sample from a Laplace distribution with unknown location parameter $\mu \in \mathbb{R}$ and known scale parameter $b = 1$, whose probability density function is given by:

$$f(x; \mu) = \frac{1}{2} e^{-|x-\mu|}, \qquad x \in \mathbb{R}.$$

Determine the maximum likelihood estimator of the location parameter μ.

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- Since $X_1, \ldots, X_n$ are independent, the likelihood function is

$$L(\mu) = \prod_{i=1}^{n} f(X_i; \mu).$$

- Substituting the density,

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{2} e^{-|X_i - \mu|}.$$

- Simplifying,

$$L(\mu) = \left(\frac{1}{2}\right)^n \exp\left(-\sum_{i=1}^{n} |X_i - \mu|\right).$$

- Taking the natural logarithm,

$$\ln L(\mu) = -n \ln 2 - \sum_{i=1}^{n} |X_i - \mu|.$$

- Since $-n \ln 2$ is a constant, maximizing the log-likelihood is equivalent to minimizing

$$S(\mu) = \sum_{i=1}^{n} |X_i - \mu|.$$

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- We need to find the value of μ that minimizes

$$S(\mu) = \sum_{i=1}^{n} |X_i - \mu|.$$

- Let the ordered sample values be

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}.$$

- Recall that, whenever $\mu \neq X_i$,

$$\frac{d}{d\mu}|X_i - \mu| = \begin{cases} -1, & \mu < X_i, \\ +1, & \mu > X_i. \end{cases}$$

- Therefore, where the derivative exists,

$$\frac{dS(\mu)}{d\mu} = -\sum_{i=1}^{n} \operatorname{sgn}(X_i - \mu).$$

- Equivalently,

$$\boxed{S'(\mu) = \#\{X_i < \mu\} - \#\{X_i > \mu\}.}$$

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- Suppose n is odd, so that
$$n = 2m + 1.$$
- When μ lies strictly between consecutive ordered observations,
$$X_{(k)} < \mu < X_{(k+1)},$$
there are k observations less than μ and $n - k$ observations greater than μ.
- Therefore,
$$S'(\mu) = k - (n - k) = 2k - n.$$
- If $k \leq m$, then
$$S'(\mu) = 2k - (2m + 1) < 0.$$
- If $k \geq m + 1$, then
$$S'(\mu) = 2k - (2m + 1) > 0.$$
- Thus, $S(\mu)$ decreases up to the middle observation and increases after it.
- Hence, the unique minimizer is
$$\boxed{\hat{\mu}_{\text{MLE}} = X_{(m+1)}.}$$

If n is odd
Take $X_{\frac{n+1}{2}}$
If n is even
Take
$\frac{X_{\frac{n}{2}} + X_{\frac{n}{2}+1}}{2}$
n: 2m+1

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- Suppose n is even, so that
$$n = 2m.$$


- When $X_{(k)} < \mu < X_{(k+1)}$, we have
$$S'(\mu) = 2k - 2m.$$
- Therefore,
$$S'(\mu) \begin{cases} < 0, & k < m, \\ = 0, & k = m, \\ > 0, & k > m. \end{cases}$$
- Thus, $S(\mu)$ decreases before the central observations, remains constant between them, and increases afterward.
- Hence, every value in the interval
$$\boxed{\hat{\mu}_{\mathrm{MLE}} \in [X_{(m)}, X_{(m+1)}]}$$
maximizes the likelihood.
- A conventional choice is the midpoint:
$$\boxed{\hat{\mu}_{\mathrm{MLE}} = \frac{X_{(m)} + X_{(m+1)}}{2}.}$$

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